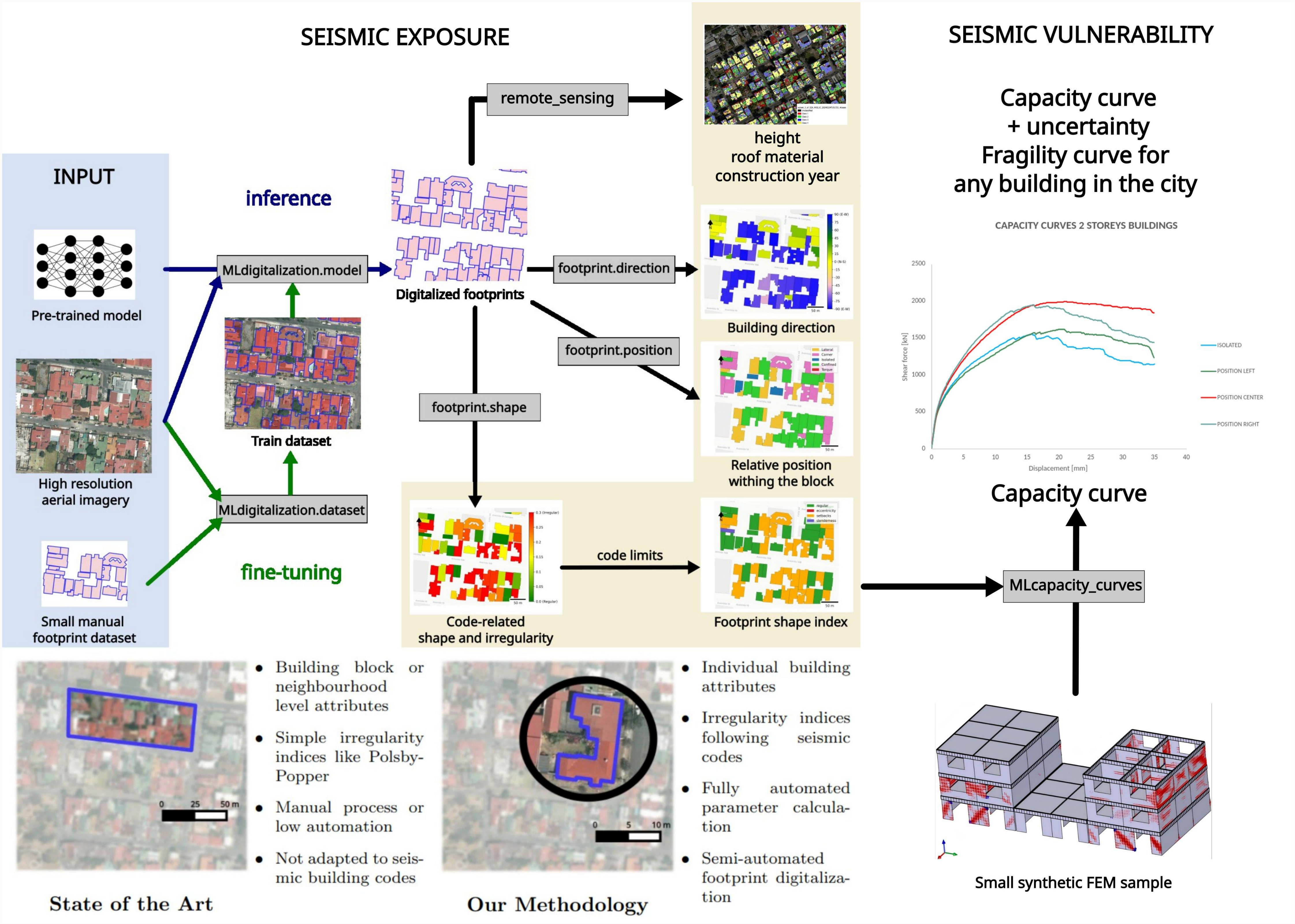




Alumno/Student: Miguel Ureña Pliego Tutor: Miguel Marchamalo Sacristán

Abstract: This thesis presents a scalable framework for seismic risk assessment in data-scarce urban areas by combining remote sensing, machine learning, and structural engineering principles. It introduces a Hybrid Twin approach that converts satellite imagery into detailed building exposure data using AI-based footprint extraction and automated geometric analysis. Seismic design codes are translated into computational rules to enable consistent estimation of structural irregularities and vulnerability. Machine learning models are used to infer building typologies from spatial features. Case studies in Latin America demonstrate significant cost reduction and practical applicability at city scale. The framework supports faster, more accessible, and more proactive disaster risk management.



	San José	Guatemala City	Santo Domingo
Traditional Workflow (USD)	355,325	416,345	402,057
Hybrid Digital Twin (USD)	54,898	41,485	45,153
Reduction (%)	84.56%	90.06%	88.76%
Cost Reduction (×)	6.5×	10.0×	8.9×